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TOWARDS THE INCORPORATION OF FLEXIBILITY, MAINTENANCE AND SAFETY IN PROCESS DESIGN

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ABSTRACT

This paper presents a method for integrating maintenance and safety aspects in the operability analysis of process systems. A stochastic process model representation is employed accounting for random process variations, equipment, failures and/or malfunctions, external unexpected events as well as different maintenance policies. The effects of events which may result in undesirable or even hazardous conditions is evaluated via a combined flexibility-reliability-safety analysis. This results in the systematic identification and ranking of critical parts of the system; implications for preventive/corrective maintenance and design optimization are discussed.

KEYWORDS

Flexibility; Reliability; Maintenance; Safety; Process Design.

INTRODUCTION

Increasing standards for product quality, safety and the enforcement of strict environmental regulations, have gradually started to motivate the need of expanding the traditional process design objectives of (short-term) profitability to include operability (flexibility, reliability, controllability, maintenance and safety) aspects in the early design stages for longer-term economic viability.
This paper is part of our ongoing research program at Imperial College. The main objective is to develop systematic techniques and tools for the integration of Flexibility, Reliability, Availability, Maintenance and Safety in process analysis and design (the FRAMS project, Thomaidis and Pistikopoulos, 1994). Some recent developments for the identification of reliability/safety "bottlenecks" in a process system are presented and remedies for enhancing operability performance are proposed based on maintenance-design optimization. A detailed case study is utilized to illustrate the key ideas of the method.

COMBINED FLEXIBILITY-RELIABILITY-SAFETY ANALYSIS

Consider the process configuration, as a part of a more complex process structure, shown in Figure 1. It comprises a tank (TK-1) for storage of a liquid solution, which feeds a flash gas polytropic compressor (C-1); the compressor is powered by a steam turbine (ST-1) backed up by an electric motor (EM-1) which takes over automatically when the steam is low (SH) or the turbine fails. A number of secondary equipment for controlling the process are also shown in Figure 1: heat exchanger (SE-1) to keep the outlet temperature of the compressor below a specified safety level, heat exchanger (SE-2) for controlling the lubricant oil temperature in the turbine and the gear box, oil filters (OF-1) and (OF-2), and control valves (V-1) and (V-2) associated with knock out drums (D-1) and (D-2), respectively. The model describing the operation of the flash gas polytropic compressor is shown in Table 1 along with process constraints, process data and data describing stochastic variations in the model. A number of external events affecting the operability of the system are also considered. Reliability (and availability if recorded) data for the equipment are given in Table 2, where a time horizon of 1000 hrs is considered. Note that although the process configuration involves 16 parts of equipment and 12 external events, its corresponding state space for operability analysis would involve approximately 268×10^6 discrete operating states (one failure mode)! The problem of how to assess the relative importance of different parts of the system (single equipment,

subsystems, external events) on the overall system performance (criticality rating analysis) can be addressed by applying the following steps:

1. Aggregation.

The system can be systematically aggregated into subsystems as follows (see Figures 1 and 2): equipment and upstream events associated with the gear box (GB-1) can be represented by subsystem (SS-2), the heat exchanger (SE-1) with its corresponding pump (P-1) and external events can be embedded in subsystem (SS-3) and so on. The resulting aggregated system consists of four subsystems (SS-1,SS-2,SS-3,SS-4), two pumps (P1-A, P1-B), the compressor and associated electrical events, the AND gates and the dummy node (LU-1). The aggregated system and the embedded subsystems are then transformed into logic functional diagrams (though the introduction of AND/OR directed diagraphs, Lapp and Powers, 1977, Iverson, 1992, as shown in Figure 2).

Note that these functional diagrams account for the degraded operating modes of a system (due to partial equipment availability) as well. For example, the power supply system is operable when either the steam turbine is available (fully powered) or the electric motor takes over (partially powered), see Table 3. Analysis from hereafter proceeds at both the aggregated and the basic subsystems level - this decoupling allows for the consideration of a very large number of equipment, events and subsystems, Rossen, 1994.

2. Flexibility-Reliability-Safety analysis.

The aggregated system results now in a state space involving 768 discrete operating states out of which only 27 are operable, as determined by a flexibility analysis step (Pistikopoulos and Mazzuchi, 1990, Straub and Grossmann, 1992, see Table 3).

Graph theory techniques are then applied to induce the minimal cut sets (Garribba et al., 1977, Billinton and Allan, 1983), and the corresponding fault-trees for the aggregated system and subsystems, as depicted in Table 4. Based on this information, a combined Flexibility-Reliability Criticality index, FRC, is defined per subsystem/equipment/event, which provides a measure of the relative importance of a "unit" within the complex process system from an overall operability perspective. The index can be computed based on the combined Flexibility-Reliability index FR (Thomaidis and Pistikopoulos, 1994) as follows (details of the mathematical manipulations are omitted here):

$$FR(t) = \sum_{k \in K^{OP}} SF_k P_k(t) \tag{1}$$

$$FRS_l(t) = \frac{\partial FR(t)}{\partial r_l(t)} \frac{\partial \rho_m(t)}{\partial r_l(t)} = \sum_{k \in K^{OP}} \left\{ SF_k \frac{\partial P_k(t)}{\partial \rho_m(t)} \frac{\partial \rho_m(t)}{\partial r_l(t)} \right\} \tag{2}$$

$$\rho_m = 1 - \prod_{c \in MC_m} \left[1 - \prod_{l \in c} r_l \right] \tag{3}$$

$$\frac{\partial P_k}{\partial \rho_m} = \prod_{j \in \{\overline{J}_k - \{m\}\}} \rho_j \prod_{j \in J_k} (1 - \rho_j), \qquad m \in \overline{J}_k$$

$$= - \prod_{j \in \overline{J}_k} \rho_j \prod_{j \in \{J_k - \{m\}\}} (1 - \rho_j), \qquad m \in J_k \tag{4}$$

$$\frac{\partial \rho_m}{\partial r_{l^*}} = \prod_{c \in \{MC_m - MC_m^*\}} \left[1 - \prod_{l \in c} r_l \right] \sum_{c^* \in MC_m^*} \left\{ \prod_{l \in \{c^* - \{l^*\}\}} r_l \prod_{c \in \{MC_m^* - \{c^*\}\}} \left[1 - \prod_{l \in c} r_l \right] \right\} \tag{5}$$

m denotes a subsystem or component/event in the aggregated system, $\rho_m(t)$ denotes the probability of failure for component/subsystem m in the aggregated level, l stands for a basic event (e.g. failure of equipment or occurrence of external event), $r_l(t)$ is a probability function of time for a basic event l. J_k ($\overline{J}_k$) is the set of operable (failed) components/subsystems in state k (aggregated level). MC_m is the set of minimal cuts associated to subsystem m, c is the minimal cut from MC_m; (l^* stands for any specific event, MC_m^* is subset of MC_m comprising all minimal cuts containing the specific event l^* and c^* is a single minimal cut from MC_m^*). SF_k denotes the stochastic flexibility index; P_k, the statistically independent state probabilities.

Note that a scaled $rFRC$ index can be defined ($0 \leq rFRC \leq 1$) by scaling all event indices according to the detected most critical one. The results of the analysis are summarised in Table 5.

3. Implications for maintenance/design

From the computed criticality indices, $\overline{FRC}$ and $rFRC$ as depicted in Table 5, it is identified that subsystem 2 (SS-2, node 1) is the most critical component at the aggregated level with the oil heat exchanger (SE-2, node 24) as the part of equipment that affects the most the overall process system's performance. If a preventive maintenance model is used to improve the availability of the oil heat exchanger (SE-2), since the most critical part of the equipment, see Table 5) to a value of 0.98511 (see Table 2), or a more reliable equipment was used, the results of this case study indicate the incremental benefits that the process achieves (i.e. its operability is improved from 0.6480 to 0.7055 with 1000 hrs of operation) as well as the new ranking order of critical parts (note that in this case the compressor becomes the most critical equipment). Note that the criticality of external events can also be assessed within this framework (see Table 5), process interactions, equipment availability and probability of events are all taken into account.

4. Implications for Safety/design

The proposed methodology can also be utilised to identify critical parts of equipment and critical operating regions from a safety point of view. This is illustrated in Figure 3, which depicts the corresponding regions of feasible operation in the state space of the two uncertain parameters (pressure ratio, flowrate). One of the process constraints, i.e. the maximum outlet temperature of the compressor, is enforced due to safety considerations. Operating states resulting in critical operation (from safety point of view) can be identified as part of the methodology described earlier and are depicted in Figure 3, cases (c) and (d) (see also Table 3). Note that the relative contribution of the various equipment/subsystems/events to the (un)safe operation of the system can then be assessed. This clearly opens the possibility for the incorporation of model-based safety analysis in integrated design tools (FRAMS).

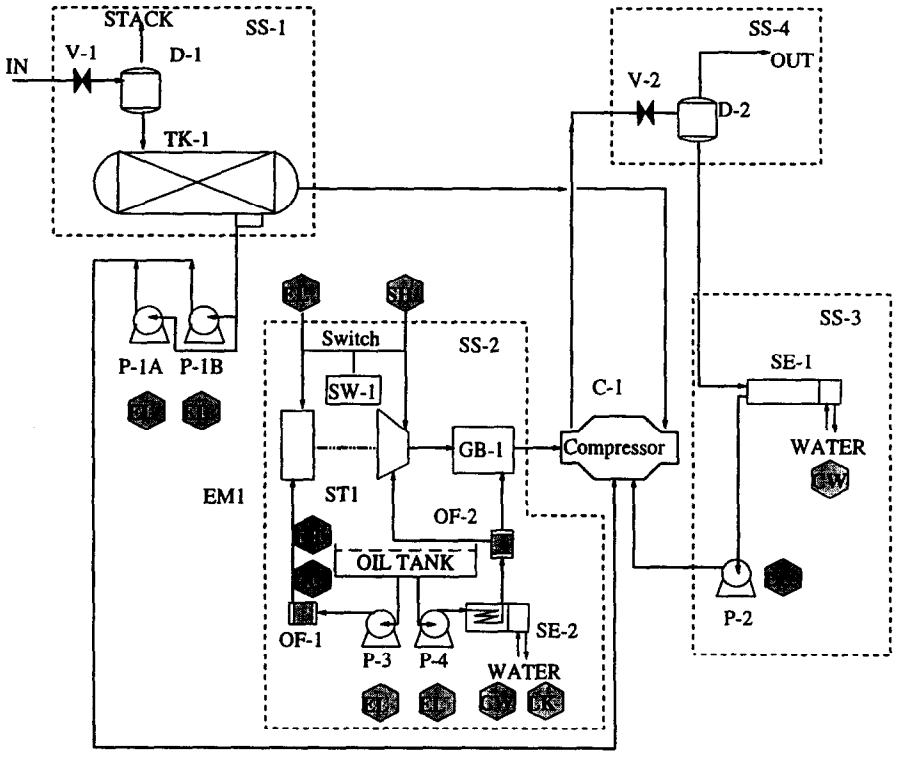


Figure 1: Process configuration

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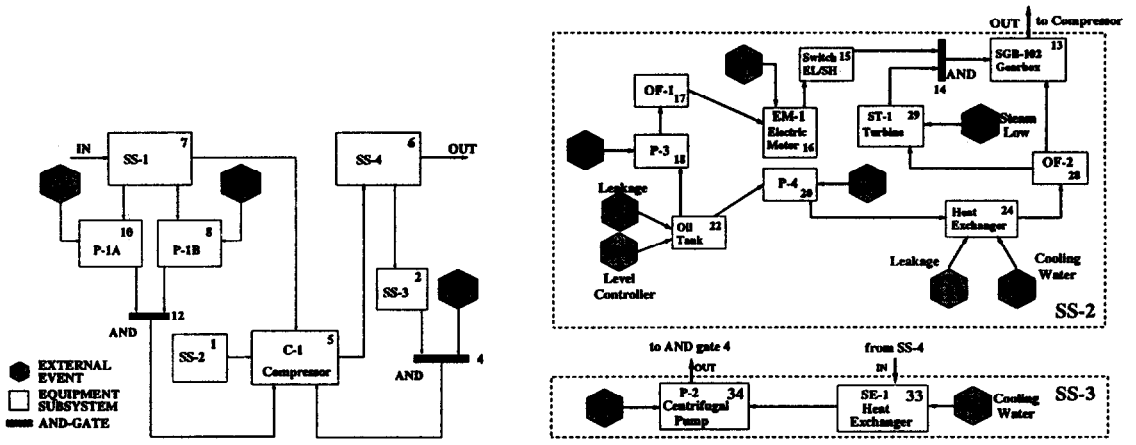


Figure 2: Functional AND/OR diagraphs (Aggregated level and subsystems)

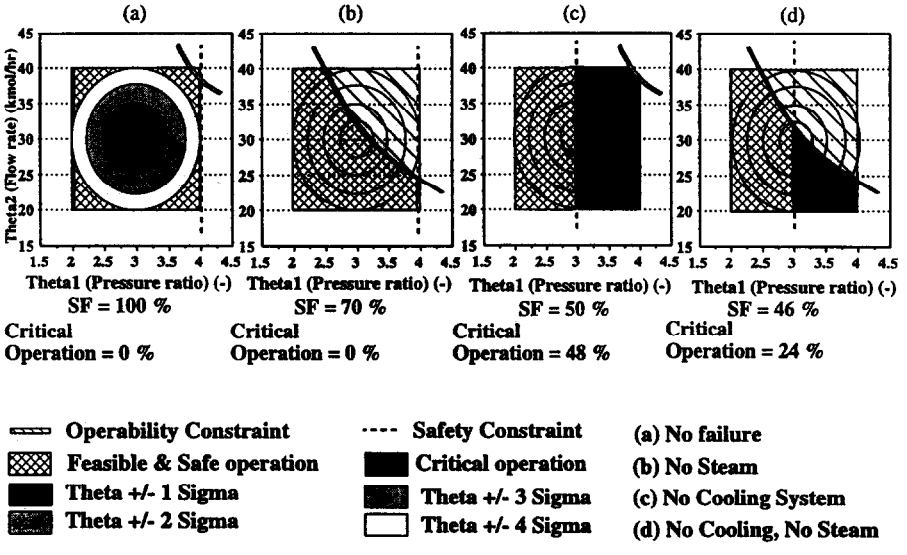


Figure 3: Safety considerations

Table 1: Process model and data

$$W_s^c = \frac{\kappa \cdot R \cdot T_{in}}{(\kappa - 1) \cdot 3600} F_{in} \left[\left(\frac{P_{out}}{P_{in}} \right)^{\frac{n-1}{n}} - 1 \right] \text{ and } T_{out} \leq T_{in} \left(\frac{P_{out}}{P_{in}} \right)^{\frac{n-1}{n}}$$

Parameter	Value	Description	Parameter	Value	Description
T_{in}	300 K	Inlet temperature	κ	1.29	c_p/c_v
n	1.398	Pres. ratio constant	η_m	0.85	Mechanical Efficiency
Maximum available horsepower			Maximum allowed outlet temperature		
Steam Turbine 60 KWatt			SS-3 Up → 440 K		
Electric Motor 36 KWatt			SS-3 Down → 410 K		
Uncertain parameters			μ	σ	range ($\mu \pm 4\sigma$)
compr. ratio	$\frac{P_{out}}{P_{in}}$		3	0.25	1 - 5
flow rate	F_{in}		30	2.50	25 - 35

Table 2: Events and corresponding probabilities

Aggregated level			Subsystems				
	Description			Description			Description	
1	SS2 Power Gen.	0.8942	13	Gear Box	0.9705	25	OilHExLeak	0.9990
2	SS3 Cooler	0.8437	14	AND gate 14	-	26	CWaterHex	0.9901
3	Utility Node 3	-	15	Electric Switch	0.9990	27	ElForElMot	0.9901
4	AND gate 4	-	16	Electric Motor	0.9950	28	OilFilter2	0.9990
5	C-1 Compressor	0.8607	17	Oil Filter 1	0.9990	29	ST-1	0.9512
6	SS-4	0.9950	18	Oil Pump 3	0.9705	30	StmForTurb	0.9802
7	SS-1	0.9950	19	Electric Event	0.9901	31	ElEvt EM-1	0.9901
8	Pump P-1A	0.9512	20	Oil Pump 4	0.9705	32	CWaterHex	0.9901
9	Electric Event	0.9901	21	Oil Leakage	0.9990	33	CoolHeatEx	0.9048
10	Pump P-1B	0.9512	22	Oil Tank	1.0000	34	Pump P-2	0.9512
11	Elctric Event	0.9901	23	Level Cntrl	0.9901	35	ElEvt P-2	0.9901
12	AND gate 12	-	24	Oil Heat Exchanger	0.9048	24*	*new Oil Heat Exchanger	**0.98511**
Time horizon 1000 hrs								

Table 3: Operating states and corresponding flexibility indices

State No	Flex. Index (%)	Crit. Oper. (%)	Subsystem No 123456789111 012	State No	Flex. Index (%)	Crit. Oper. (%)	Subsystem No 123456789111 012
1	100	-	111111111111	34	50	48	101111111011
2	70	-	D11111111111	40	70	-	D11111111101
3	50	48	101111111111	41	50	48	101111111101
7	100	-	111111101111	47	100	24	111111111001
8	100	-	111111110111	74	45	24	D01111110111
9	100	-	111111111011	84	70	-	D11111100111
10	100	-	111111111101	85	50	48	101111100111
12	45	24	D01111111111	90	45	24	D01111111011
22	70	-	D11111101111	112	45	24	D01111111101
23	50	24	101111101111	133	70	-	D11111111001
27	70	-	D11111110111	134	50	48	101111111001
28	50	48	101111110111	180	45	24	D01111100111
32	100	-	111111100111	273	45	24	D01111111001
33	70	-	P11111111011	Total States 768			

Aggregated System	
Component/System	Operating modes
1	1: fully operable, D: degraded, 0: failed
2,5,6,7 8,9,10,11	1: operable, 0: failed
3,4,12	none (utility nodes)

Table 4: Minimal cut sets

	Minimal Cut Sets
Aggregated	{6},{7},{1},{5},{11 9},{11 8},{10 8},{10 9}
Subsystem SS-2	{13},{24},{27},{21},{23},{20},{25},{26},{28}, {29 31},{29 17},{29 19},{29 18},{29 16},{29 15}, {30 15},{30 31},{30 17},{30 19},{30 18},{30 16}
SS-3	{32},{33},{34},{35}

Table 5: Summary of results

Before modification				After modification			
R(1000) = 0.7113 FR(1000)= 0.6480				R(1000) = 0.7744 FR(1000)= 0.7055			
Node	Description	$\frac{\partial FR(t)}{\partial r_i(t)}$	Scaled	Node	Description	$\frac{\partial FR(t)}{\partial r_i(t)}$	Scaled
Parts of equipment at the Aggregated level							
1	SS-2	-77.78581	1.00000	5	C-1	-80.80826	1.00000
5	C-1	-74.22337	0.95420	1	SS-2	-77.78581	0.96260
6	SS-4	-64.20535	0.82541	6	SS-4	-69.90147	0.86503
7	SS-1	-64.20535	0.82541	7	SS-1	-69.90147	0.86503
2	SS-3	-34.80459	0.44744	2	SS-3	-37.89236	0.46892
8	P1-A	- 3.69590	0.04751	8	P1-A	- 4.02379	0.04979
10	P1-B	- 3.69590	0.04751	10	P1-B	- 4.02379	0.04979
Events at the Aggregated level							
9,11	Electric	- 3.55098	0.04565	9,11		- 3.86601	0.04784
Parts of equipment at subsystems level							
24	OilHEx	-68.23095	0.87716	13	GearBox	-69.26215	0.85712
13	GearBox	-63.61813	0.81786	20	P-4	-69.26215	0.85712
20	P-4	-63.61813	0.81786	24	OilHEx	-68.23095	0.84436
28	OF-2	-61.79968	0.79449	28	OF-2	-67.28237	0.83262
33	CoolHEx	-32.45159	0.41719	33	CoolHEx	-35.33060	0.43722
34	P-2	-30.86890	0.39684	34	P-2	-33.60751	0.41589
18	P-3	- 4.23855	0.05449	18	P-3	- 4.61458	0.05711
15	Switch	- 4.23366	0.05443	16	El.Motor	- 4.61000	0.05705
16	El.Motor	- 4.23434	0.05444	15	Switch	- 4.60926	0.05704
17	OF-1	- 4.23366	0.05443	17	OF-1	- 4.60926	0.05704
29	SteamTurb	- 3.48790	0.04484	29	SteamTurb	- 3.79734	0.04699
Events at subsystems level							
23	Lev.Cntrl	-62.35839	0.80167	23	Lev.Cntrl	-67.89065	0.84014
26	CoolWater	-62.35839	0.80167	26	CoolWater	-67.89065	0.84014
27	Electric	-62.35839	0.80167	27	Electric	-67.89065	0.84014
21	Leakage	-61.79968	0.79449	21	Leakage	-67.28237	0.83262
25	Leakage	-61.79968	0.79449	25	Leakage	-67.28237	0.83262
32	CoolWater	-29.65852	0.38128	32	CoolWater	-32.28974	0.39958
35	Electric	-29.65852	0.38128	35	Electric	-32.28974	0.39958
19	Electric	- 4.23519	0.05445	19	Electric	- 4.61092	0.05706
31	Electric	- 4.23519	0.05445	31	Electric	- 4.61092	0.05706
30	Steam	- 3.48593	0.04481	30	Steam	- 3.79519	0.04697
Utility nodes at the Aggregated & Subsystem level: all zero							